

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Improves code readability and consistency.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, model, and backend.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Improves code understanding and maintenance.
4	Function / Method Names	Functions are descriptively named	Yes	Makes the code easier to understand and reuse.
5	Code Comments	Inline and block comments explain logic	Yes	Helps developers understand the implementation.
6	Modular Design	Code is split into reusable functions/modules	Yes	Backend logic is organized into separate functions.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reduces complexity and improves maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Ensures stable and reliable application behavior.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Backend (app.py)	Python, Flask	Handles routing and prediction requests	High
2	Machine Learning Model (model.pkl)	Scikit-learn, Pickle	Used for crop prediction	High
3	Data Preprocessing Module	Python, Pandas, NumPy	Cleans and prepares input data	High
4	Frontend Components	HTML, CSS, JavaScript	User interface for data input and result display	Medium
5	Prediction Function	Python	Generates crop recommendations based on user inputs	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Code is well-organized with a clear folder structure.
Readability	5	Meaningful variable names, functions, and comments improve readability.
Reusability	5	Modular design and reusable prediction logic support future enhancements.
Documentation / Comments	4	Most important sections are documented with comments.
Overall Score	4.8	The codebase is structured, maintainable, and suitable for future development.